

PRAKRIT RAMACHANDRAN

Mechanical Engineer — Robotics, Mechatronics, CAD, FEA & Manufacturing

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SUMMARY

Mechanical engineering student (B.S., University of Washington, expected 2028) who builds mechatronic hardware end-to-end: CAD, FEA, closed-loop control, prototyping, and validation. Led the mechanical design and fabrication of a competition drone (SolidWorks FEA, custom gripping end-effector), built a precision CoreXY motion system with GRBL control, and shipped production web and iOS software as co-founder of PropLane (proplane.ai). Skilled in SolidWorks, Fusion 360, Onshape, AutoCAD, GD&T, DFM, 3D printing, CNC machining, laser cutting, and welding. Eager to build cutting-edge robotics and mechanical hardware on a fast-moving team.

EDUCATION

University of Washington — B.S. Mechanical Engineering

Expected 2028

Seattle, WA

2024 – Present

Relevant Coursework: System Dynamics & Controls, Kinematics, Statics, Mechanics of Materials, Materials Science, Thermodynamics, Manufacturing Processes, CAD & Engineering Graphics

EXPERIENCE

Mechanical Engineer — Husky Robotics, Drone Subteam

Seattle, WA

Collegiate robotics team

Jan 2025 – Jan 2026

- Led mechanical design and fabrication of a competition drone through a 10-week build cycle, fabricating parts in carbon fiber, machined aluminum, and 3D-printed polymers to meet payload, weight, and structural constraints.
- Ran SolidWorks FEA to optimize airframe geometry, cutting mass ~20% while preserving structural integrity under load.
- Designed a multi-orientation gripping end-effector for reliable payload capture during dynamic flight, refining it over ~5 prototype iterations validated in flight testing.
- Produced detailed CAD assemblies and manufacturing drawings and integrated mechanical systems with the electrical and software subteams in formal design reviews.

Co-Founder & Software Lead — PropLane (proplane.ai)

Seattle, WA

AI-powered property management platform

2026 – Present

- Co-founded and led a 3-person team to build an AI-powered property-management platform from the ground up — the AI generates leases, powers core web-app features, and answers resident questions — shipping web and iOS apps to production (React Native/Expo, Next.js, Supabase, Stripe).
- Architected and integrated multiple third-party systems — payments, tenant screening, and SMS — into one full-stack product, owning the build end-to-end across frontend, backend, and infrastructure.
- Personally own and manage two rental properties (9- and 10-bedroom) generating over \$10K in monthly recurring revenue, and now onboarding external properties onto the platform with a path to licensing it to other property managers.

PROJECTS

2-Axis Pen Plotter — CoreXY Motion System

- Designed and built a precision 2-axis CoreXY motion system with belt-driven actuation and stepper motors, engineering 20+ custom structural and motion components to render complex 2D geometries.
- Tuned frame rigidity, belt tension, and motion parameters to cut backlash and reduce vibration, measurably improving positional accuracy and repeatability.
- Implemented an Arduino control stack running GRBL firmware for synchronized multi-axis motion, and calibrated the system for repeatability through iterative testing.

Marble Coaster — Continuous-Motion Mechanism

- Designed and fabricated a continuous-motion marble coaster from concept through assembly, modeling 10+ custom components including a custom helical lift and a 3D-printed gearbox that recirculates marbles automatically.
- Sized track curvature and drop heights from energy-loss (friction and rolling-resistance) calculations and applied kinematics and energy-conservation principles to deliver reliable, derailment-free runs, refining geometry through repeated physical testing.

SKILLS & INTERESTS

Robotics & Mechatronics: Closed-Loop / PID Control, Motion Control Systems, Stepper/Servo Actuation, GRBL, Sensor Feedback (Gyro, Odometry), Systems Integration

CAD & Analysis: SolidWorks, Fusion 360, Onshape, AutoCAD, CAD Assemblies & Engineering Drawings, GD&T & Tolerance Analysis, FEA / SolidWorks Simulation, Stress & Strain Analysis, Design for Manufacturing (DFM), Bills of Materials (BOM)

Prototyping & Fabrication: 3D Printing (FDM), CNC Machining, Laser Cutting, Welding, Rapid Prototyping, Design for Assembly (DFA), Tolerancing

Electronics & Embedded: Arduino, Raspberry Pi, Motor Drivers & Controllers, Soldering & Wiring Harnesses, PCB Layout (basic), Embedded Motion Control

Programming: Python, TypeScript / JavaScript, Java, MATLAB / Simulink, SQL

Full-Stack & AI Development: React Native / Expo, Next.js, Node.js, REST APIs, Supabase (PostgreSQL), Stripe, Git; agentic / AI-assisted coding with LLM APIs and prompt engineering

Business & Go-to-Market: Customer Acquisition, Cold Calling & Outreach, Property Management, Market Research

Interests: Robotics & mechatronics, materials & manufacturing processes, structural design & FEA, and motion systems; hands-on making with 3D printing, CNC machining, and laser cutting.

Languages: English, Hindi